University Of Engineering and Technology, Lahore
Computer Engineering Department

Lab Evaluation:

Course Name: Data Structures & Algorithm	**Course Code:** CMPE-212L
Assignment Type: Open Ended Lab	**Dated:** 5th March 2024
Semester: 4th	**Session:** 2022
Lab/Project/Assignment #: 7	**CLOs to be covered:** CLO 2
Lab Title: Stack Application	**Teacher Name:** Engr. Afeef Obaid

CLO 2	Design solution for problems using appropriate data structures and algorithm					
Levels (Marks)	**Level 1**	**Level 2**	**Level 3**	**Level 4**	**Level 5**	**Level 6**
(10)						
					Total	**/10**

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	**9-10**	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	**7-8**	L2	Submitted the lab tasks but have good understanding
Good	**5-6**	L3	Submitted the lab tasks but have weak understanding.
Basic	**3-4**	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	**1-2**	L5	Submitted only one lab task.
Not Acceptable	**0**	L6	Did not attempt

LAB No. 7

Lab Goals/Objectives:

- Applications of Stack
- Bracket Checker
- Multiple Bracket Checker

Equipment Required: Computer and Dev C++ Compiler

Theory/Relevant Material: Stack Data Structure, Link List and Array

Lab Tasks:

- Being programmer, we know most of the times syntactic errors result from misplaced and missed brackets. Therefore, a program is required that takes a sequence of brackets from user and validates the file for usage of correct bracket. It should check following things:
  - **a)** Bracket sequence can contain [, (,{ i.e all known brackets. Devise an algorithm to check if brackets are used correctly or not.
  - **b)** Identify what is the error type: misplaced bracket, missed bracket
  - **c)** Identify which bracket caused error i.e., the bracket's position or index number